

Vansh Sharma

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Education

B.S. in Computer Science - **California State University East Bay, Hayward CA**

January 2021 - May 2025

- **Cumulative GPA: 3.9, Dean's Honors List** for 6 semesters, **Summa Cum Laude Awardee** for graduating with highest level of academic distinction.
- Coursework: Data Structures & Algorithm, Software Engineering, Operating System, Computer Networks, Cybersecurity, Introduction to NLP with DL

Skills

- Programming and Scripting Language: C, C++, Python, Java, JavaScript, TypeScript, PHP, XML, HTML/CSS, MIPS, x86 Assembly Language
- Libraries and Frameworks: React, Node.js, Express.js, NestJS, MERN Stack, GraphQL, JUnit, Hasura, Scikit-learn, Pandas, NumPy, Matplotlib, Chart.js
- Database and Backend Development: REST API, WebSocket, Firebase, MongoDB, PostgreSQL, Prisma, Apollo Client, Multi-threading, Parallel Processing
- Software Architecture and DevOps: Domain-Driven Design, Design Patterns, SDLC, Amazon Web Services, Google Cloud Platform, Heroku, Supabase
- Tools and Security: Git, Postman, Eclipse, VSCode, Firecamp, Unix, Linux, Kernel Development, Cryptography, Cybersecurity Protocols, Vibe Coding

Work Experience

Senior Software Engineer - **Owting, Los Angeles CA**

July 2025 - Present

- Engineered a cross-functional profile verification feature to improve user safety by deploying a **Python-based Cloud Run** microservice on Google Cloud Platform (GCP) built within a custom **Docker** container. Leveraged the **deepface** open-source library with the **ArcFace** deep learning model for accurate facial recognition to achieve a 92% match accuracy and used **Google Vision API** to perform liveness and spoofing checks. The end-to-end system-level pipeline was triggered via **Pub/Sub**, performed profile verification checks, which then processed the results and notified the user via a Postgres event trigger on the subsequent database update. Leveraged a separate microservice to dispatch push notification to fully automate the security workflow.
- Architected and programmed an automated referral rewards system to incentivize platform growth, inserting referral data into **BigQuery** for analysis. Implemented an asynchronous event-driven workflow where a **Python** function is triggered upon new user registration and queries BigQuery to determine badge eligibility for the referrer. Upon a certain number of successful referrals, the system automatically calls a rewards service to grant the badge to the user via MongoDB, which then uses a **MongoDB Atlas Trigger** to call the notification dispatcher to send a real-time push notification.

Software Engineer - **Owting, Los Angeles CA**

August 2024 - July 2025

- Architected and implemented real-time features like typing indicators, soft message delete, edit message and message pinning within the chat-messaging **microservice** using **WebSockets**, **Nest.js**, **Typescript** and **asynchronous event-driven APIs** connected to **MongoDB** via **Prisma ORM**, optimizing message transmission to achieve a 30% reduction in perceived latency and significantly enhance chat functionality and user experience.
- Designed and developed a secure media upload functionality by implementing streaming data transfer to **Google Cloud Storage**. Developed REST APIs to generate GCS signed URLs for media upload, track status, trigger downstream processing via Pub/Sub and rendering image using **CDN** signed URL. Configured buckets and service accounts to enable multimedia sharing and ensure scalable storage infrastructure capable of handling >1TB of data.
- Built **RESTful APIs** using **Nest.js** framework for the data processor microservice, enabling CRUD operations on users' general and date-specific advanced availability via **PostgreSQL**. Improved API response time by 15% through query optimization, ensuring data integrity and accurate availability tracking.
- Created efficient RESTful API endpoints allowing users to generate multiple distinct activity events from a single request. Integrated **GraphQL** using **Apollo Client** for data querying, worked with UI developers to ensure front-end integration which simplified user workflow for event creation by 35%.
- Developed features across distinct **microservices**, while adhering to Domain-Driven Design to enhance system modularity, testability, and scalability.

Software Engineer Intern - **Royaltutors, Remote, USA**

May 2024 - August 2024

- Designed and developed a computer-assisted learning platform to deliver educational content and facilitate interactive learning experience. Developed a student panel to practice questions and take tests, and an admin panel to create practice questions and sample tests which are stored in **PostgreSQL**.
- Implemented real-time analytics feature to compute and store student performance data like accuracy and subject proficiency. Used **REST APIs** to fetch these data on student dashboard, which used **Chart.js** to render dynamic visualizations to increase student interaction with progress-tracking tools.
- Integrated the Stripe payment gateway into the platform, which unlocked premium features, like taking sample tests or access reading material, for users upon purchase. This led to a 25% increase in revenue over the next 3 months. The techstack included: **PHP, JavaScript, PostgreSQL, HTML/CSS**.

Software Engineer Intern - **KidSTREAM, Los Angeles CA**

May 2023 - August 2023

- Collaborated with the Extended Reality (XR) lead and a multidisciplinary team to develop immersive and interactive experiences for virtual exhibits and educational programs using XR technologies like 8th Wall. Integrated 3D assets, animations, and dynamic interactions for a seamless user experience.
- Engineered 4 XR experiences: [FireFighter Selfie](#), [AR PhotoBooth](#), [FireFighter](#), [AR Room](#) for virtual exhibits using **HTML, CSS and JavaScript**.

Projects

- Text-Messaging Application: Collaborated with 3 developers to program a **Java based Text-Messaging Software** with seamlessly integrated client-server architecture and Graphical User Interface. The system supports private and group messaging and online status tracking. The admin role can view message logs. Application uses TCP/IP protocol for message transmission over the internet. Reduced the latency by 25% using parallel processing.
- Notes Application: Developed a notes taking application using **MERN stack**, implementing user authentication with password hashing and JWT-based session management. Created RESTful APIs to enable user-specific CRUD operations, with strict token validation to ensure secure access to notes.
- Boggle: Developed a Boggle game in **C++** leveraging advanced graph algorithms and recursion. Created a dynamic dictionary structure to validate words and implemented a depth-first search (DFS) algorithm to traverse a letter matrix, efficiently generating all valid word combinations in the matrix.
- Linux Kernel Module (Operating Systems CS 421): Designed a **multi-threaded Linux kernel** in **C**. Programmed CPU scheduling algorithms such as First Come First Serve, Round Robin, and Shortest Time Remaining First which resulted in 50% reduction in average latency and a 35% increase in throughput. Conducted performance testing and scalability analysis under up to 50 concurrent threads, ensuring system reliability and stability.